Rotatable Evaporative Windcatcher Integrated with Solar Chimney

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I. Background

Over the past decades, significant advancements have been made in high-performance, energy-efficient building technologies. Overall, these energy-related technologies fall into two main categories [1]. The first includes building equipment, such as heating, ventilation, and air conditioning (HVAC) systems, which actively consume energy to enhance occupant comfort and indoor air quality. Recent innovations in this area include solar-driven fifth-generation district heat pumps [2] or membrane-based dehumidifiers [3]. The second category is the building envelope, comprising walls, windows, etc., which insulates interiors from external conditions to reduce heat transfer and consequently required cooling or heating loads. Recent advancements in this area include electrochromic windows [4], vacuum insulation panels [5], and elastic building envelopes [6].

While great advancements with promising results are made in both categories to improve buildings' total energy efficiency, their high final costs remain a significant barrier to their widespread adoption for both new and existing buildings. While the incremental cost of constructing energy-efficient buildings may become minimal with proper initial planning and training, retrofitting existing buildings poses greater challenges as many households struggle with the upfront costs, and retrofitting often requires substantial upgrades to existing structures and infrastructure, leading to higher remodeling costs than those associated with new builds [7]. This financial burden is particularly severe for low-to-moderate income (LMI) communities, who often face the greatest need for energy-saving technologies but have limited means to afford them [8]. Therefore, there is an urgent need to develop affordable, energy-efficient solutions that reduce costs to stakeholders (building owners, manufacturers, etc.) while maintaining essential occupant indoor comfort. These technologies should be designed for easy production and integration into existing buildings to address the unique challenges faced by LMI groups.

II. Problem Statement

Following the above challenge, this work specifically addresses the rising cooling costs in U.S. residential buildings, a growing concern as global warming and the urban heat island (UHI) effect extend the need for cooling across more regions and for longer periods [9]. Recent studies indicate that climate change could drive a 20-35% increase in cooling energy consumption in the U.S. by the end of the century, highlighting the critical need for efficient cooling systems to ensure building affordability [10]. The proposed system is designed as an add-on device to existing building envelopes with minimal site disturbances, targeting the challenge of retrofitting current buildings affordably for LMI populations. While primarily intended for existing buildings, the system can easily be integrated into new construction designs as well. The system is economically suited to dry climates with cooling needs, either year-round or seasonally, and wind speeds of 4 m/s or above as further detailed in section IV.

III. Solution

The proposed solution that targets the above problem is called Rotatable Evaporative Windcatcher Integrated with Solar Chimney (REWISC). To properly elaborate the solution, short explanations of windcatchers and solar chimneys, and their integration are firstly included, followed by the problems and current gaps that cause the technology to fall behind and not be integrated into real buildings. In the next section, our solution (REWISC) which properly targets the problem while handling the current challenges is illustrated and explained in detail.

Windcatchers. Prior to the development of mechanical cooling systems, windcatchers were widely used for cooling and ventilation in hot climates, particularly in the Middle East where they originated. A windcatcher is essentially a rooftop tower that captures wind at higher elevations and channels it into the buildings' inner environment [11]. Fig. 1 (A) schematically illustrates this wind-driven ventilation system. By the movement of external wind at the roof level, a positive pressure on the windward side of the structure (wind force) and at the same time, negative pressure on the leeward side (stack effect buoyancy force) are produced. This pressure difference on the two sides drives fresh air into the building while expelling warm air [12]. In many traditional applications, windcatchers were combined with

evaporative cooling, where incoming air flowed over a water pond or underground canal to get cooled, enhancing the cooling process without requiring conventional mechanical systems [13].

The traditional windcatchers are classified into four main groups based on the number of their open sides including a) one-sided windcatchers, which take the wind from the top opening, channel it inside, and afterward the air exits from another opening of the building [13]. They were typically utilized in the areas where the air blows in one specific direction; b) two-sided windcatchers, which have two openings on two opposite sides so that one opening is used as an entrance for fresh air and another extracts the warm air [14]; c) four-sided windcatchers, designed to capture the prevailing wind from four main directions. [15]; and finally, d) multiple-sided windcatchers, which were suitable for areas with no specific direction for wind [84]. Modern windcatchers (mainly constructed in Middle Eastern and European countries, especially the UK) utilize traditional windcatcher technologies and adopt them with advanced building principles such as adjustable dampers, horizontal louvers, etc. [12].

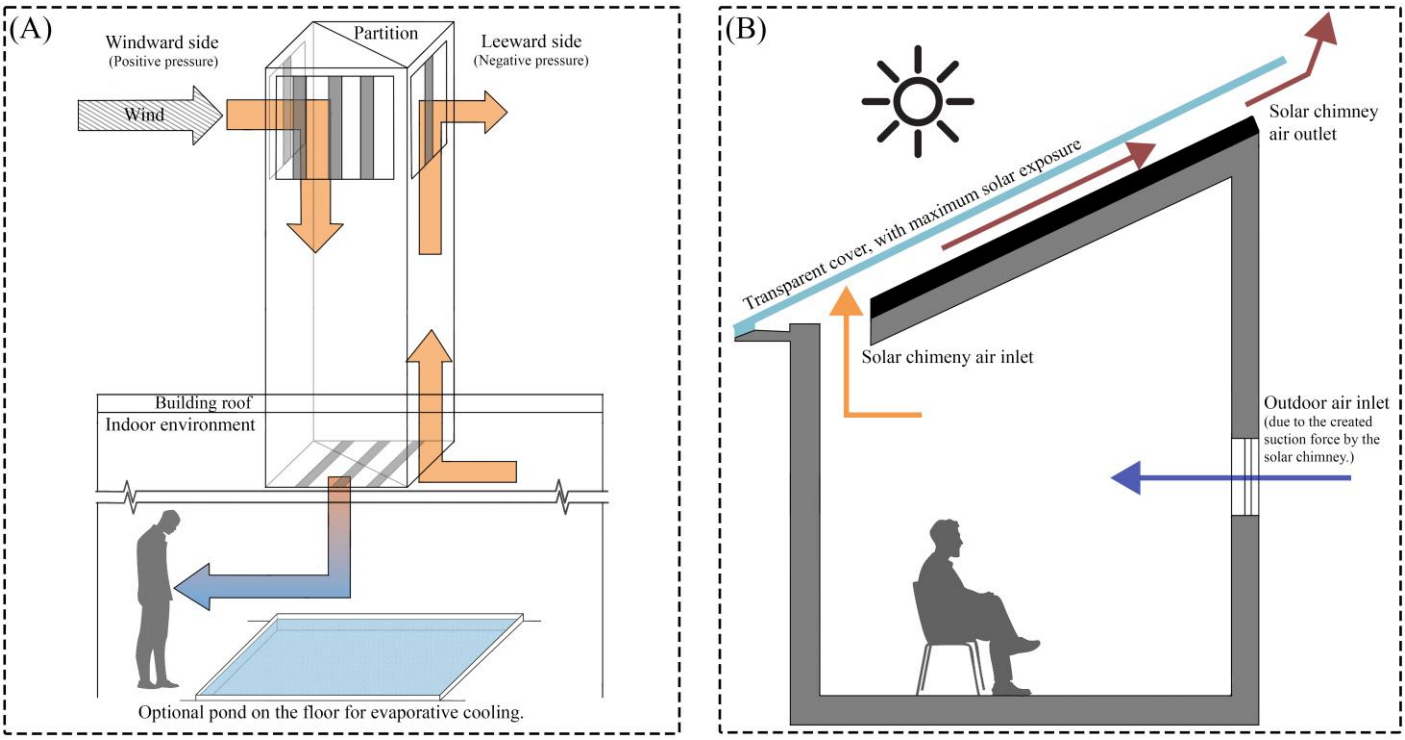


Fig. 1. Schematic illustration of (A) windcatchers; and (B) solar chimneys (adapted from [16] and modified)

Solar Chimneys. As shown in Fig. 1 (B), solar chimneys use maximum solar exposure to create sufficient temperature and consequently density difference between their inlet and outlet, resulting in a stack effect that pushes indoor air toward the outlet [17]. This stack effect generates a suction force which draws outside air into the building through window openings, enabling natural ventilation even on windless days. Various types of solar chimneys, including roof-mounted models and Trombe walls, are widely used to promote natural ventilation in sunny climates [18, 19].

Windcatchers and Solar Chimneys Integration. Limited studies have preliminarily explored integrating solar chimneys with windcatchers to harness the benefits of both systems: the windcatcher's ability to draw outdoor air inside, and the solar chimney's suction to expel warm air [20-22]. However, this integration has not advanced beyond a simple attachment of the two components, failing to address real-world challenges, which limits its practical feasibility. To overcome these issues, we propose a new design, the Rotatable Evaporative Windcatcher Integrated with Solar Chimney (REWISC), to be added to existing building structures for enhanced cooling in summer or year-round. Fig. 2 illustrates this proposed solution, incorporating the below techniques to optimize its performance:

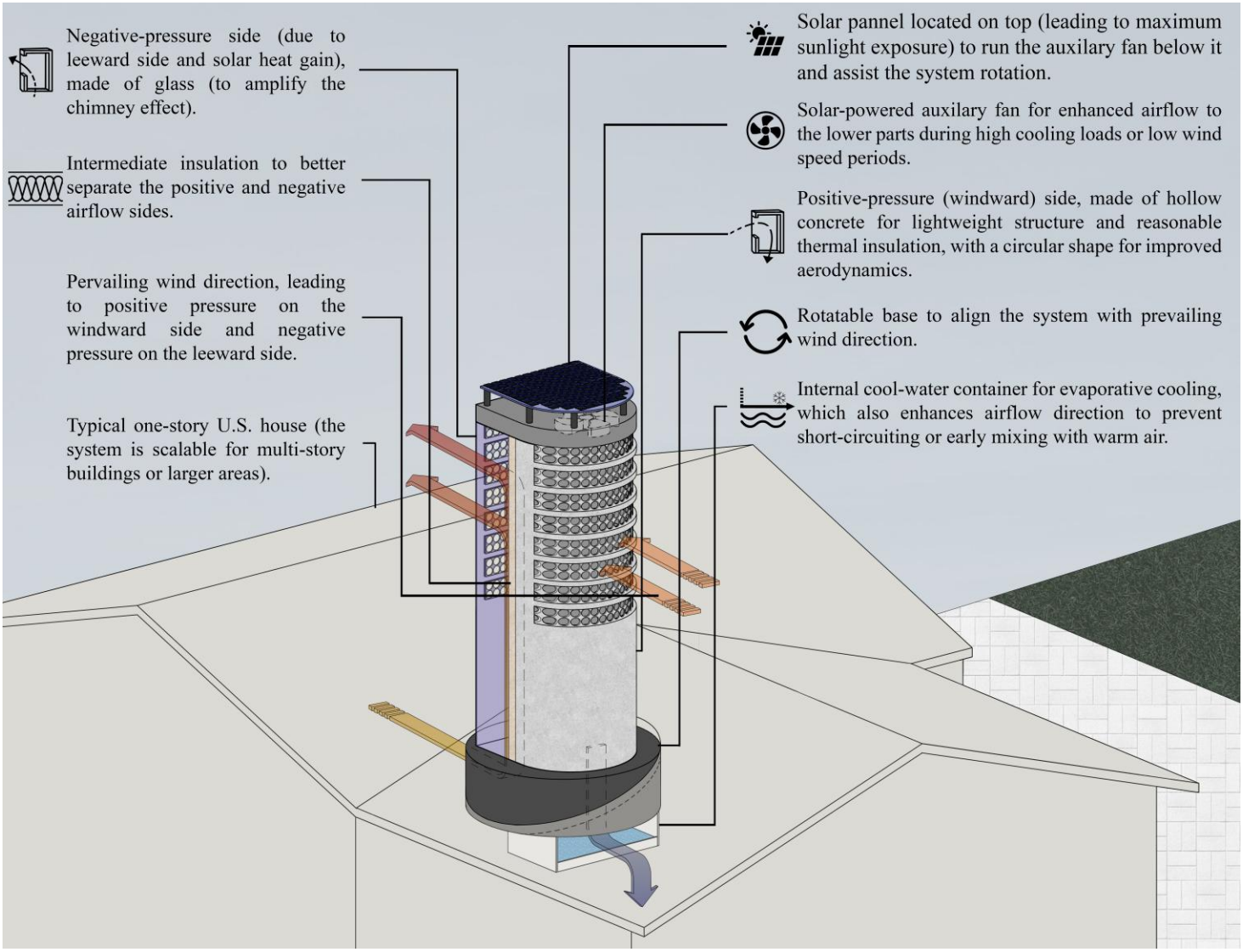


Fig. 2. Schematic diagram of the proposed solution

1. Rotatable Base for Multi-Directional Winds. In most American cities, prevailing wind direction varies throughout the year, with no single dominant direction. A stationary chimney and windcatcher setup could be counterproductive in these conditions, potentially drawing in hot air from the chimney side and exhausting cooler air from the windcatcher. To address this, we designed a simple rotatable base at the roof-system junction, allowing the structure to align with the dominant heat direction based on local meteorological data. This rotation maximizes airflow efficiency without adding much complexity or extra load on the roof. It can be powered by the solar panel mounted on top of the device, with battery backup for periods without sunlight.
2. Aerodynamic Design of Positive and Negative Pressure Zones. The positive pressure intake is designed to be circular, maximizing wind capture and directing it into the system. This section uses lightweight, thermally resistant materials like hollow concrete to minimize roof load. The other side, designed for efficient air removal, is rectangular with voids positioned exclusively on the leeward (negative pressure) side. Glass is used here to absorb sunlight, increasing buoyancy and accelerating upward air movement. An insulating layer separates the intake and exhaust zones to minimize heat transfer.
3. Filtered Medium-Sized Voids for Airflow Control. Instead of large openings, which can allow debris (e.g., leaves) into the system, multiple medium-sized, netted voids are used on both sides to filter particles while ensuring sufficient airflow. These voids can be closed in winter or when the system is not in use, preventing heat loss from the system during colder months.
4. Auxiliary Solar-Powered Fan. A fan, powered by a solar panel located on its top, assists with airflow when natural wind flow is insufficient or when higher air circulation is needed. Placing the solar panel at this height ensures

consistent sun exposure, unobstructed by surrounding buildings or trees. It can also supply power to the rotatable base as discussed earlier.

5. Evaporative Cooling with a Water Container. To replicate the effect of water ponds in traditional windcatchers, a $1m \times 2m$ water container is located at the end of the positive pressure side. Since the water container is placed indoors, shielded from direct sunlight and heat, incoming air is cooled before entering the building. The container also directs the incoming air toward the side opposite the negative pressure zone, which prevents short-circuiting (i.e., ensuring that cool air flows through the house, gradually warms, and then exits through the negative pressure side.)

IV. Technology-to-Market Plan

Optimal Installation Regions. Experimentally, windcatchers perform best in locations with wind speeds of 4 m/s or higher [23]. Fig. 3 (A) identifies U.S. regions meeting this wind speed threshold during the summer [24], with July 2024 chosen as a representative month. Additionally, since the system relies on evaporative cooling, it is most effective in areas with significant cooling demands (year-round or in summer) and dry climates. Using the ASHRAE geographical weather map [25], suitable zones would be ASHRAE climate zones 0B, 1B, 2B, 3B, and 4B (identified and highlighted in Fig. 3 (B)). Overlaying these two criteria reveals regions where installing this system would be both efficient and cost-effective for residents, as illustrated in Fig. 3 (C). The final map shows that, with appropriate promotion and development, this technology holds substantial market potential for installation across the U.S.

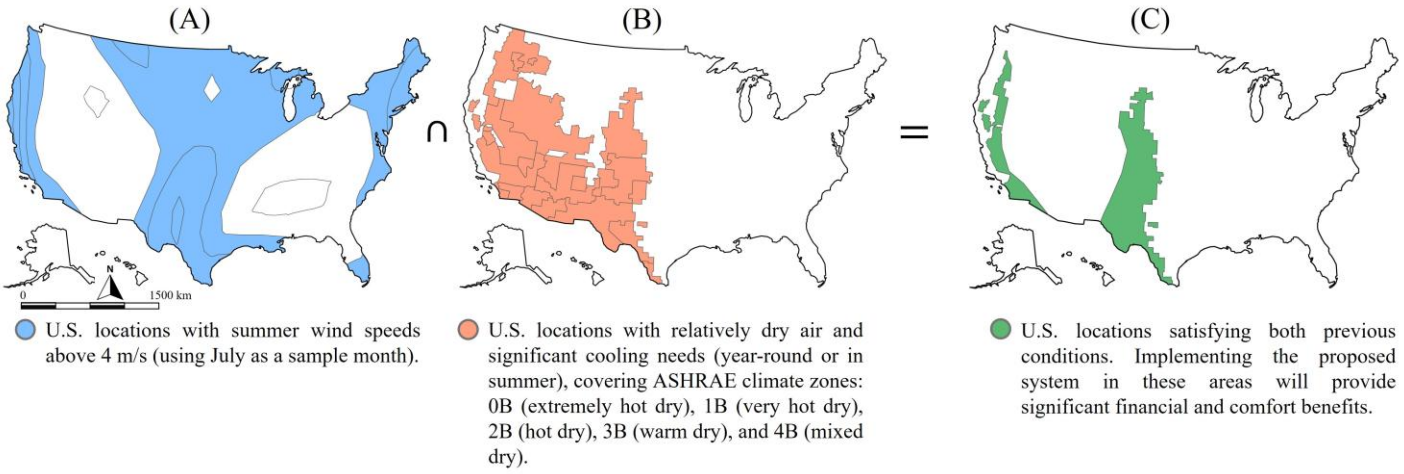


Fig. 3. US regions that (A) satisfy the wind requirement (wind speed of 4 m/s or higher); (B) satisfy the weather requirement (dry air and requiring significant cooling load year-around or in summer); and (C) satisfy both wind and weather requirements.

From Concept to Installation. To integrate this idea into the design of real buildings, the following process is required. 1) Among the potential regions identified in Fig. 3 (C), a specific location must be determined, where installation of this system, with its dimensions and specifications, is permitted by regional or state governments (or any other agencies that set or adapt the national buildings codes in that city). 2) A site analysis is then required to evaluate site-specific factors that determine system specifications, such as its height (avoiding overly tall systems in low-wind areas or too short systems in high-altitude wind zones) and whether a rotating mechanism is required (some locations may have consistent wind directions, making rotation unnecessary), etc. The results can be used to select the final system from predefined options (such as short, medium, or tall and rotatable or non-rotatable systems). This approach balances cost-efficiency (avoiding custom designs for each house) with maximum effectiveness (by tailoring to local conditions). Key stakeholders in this stage are firstly, city (or state) construction firms that can allocate personnel, with enough knowledge of local conditions, for this brief (1–2 days) analysis. Secondly, local communities and representatives are required to engage with residents and homeowners to inform them of the system's benefits, ensuring their acceptance and cooperation. 3) Based on the site analysis results, the system can be constructed or selected from prefabricated modules and transported for installation. Due to the system's simple structure, installation can be completed in a day, and trained installers, educated about the system installation, may be joined by local workers to support community involvement.

Cost/Benefit Analysis. The estimated manufacturing costs of the proposed solution are detailed in Table 1, with a final total cost of $1,045.90. Due to the climate-dependent nature of the system, precise energy savings are challenging to calculate [26]. However, studies indicate that in warm, arid climates, the integrated windcatcher and solar chimney can lead to substantial cooling energy savings of up to 75% during the peak cooling seasons [20]. This analysis assumes that an A/C unit remains necessary for extremely hot and windless conditions, and complete replacement may not be feasible. A typical single-story house in Amarillo, TX (as an example of cities in regions identified in Fig. 3 (C)) pays an average of $48.62 per month for A/C during the summer [27]. With an estimated 75% savings, the system could reduce monthly A/C expenses by approximately $36.46 ($0.75 \times \$48.62 = \$36.46$). For a four-month cooling season, the payback period for the initial investment would be calculated as:

$$Payback = \frac{\$1045.9\ Initial\ Const}{\left(4\ months \times \frac{\$36.46}{month}\right)\ Annual\ Saving} = 7\ years.$$

This relatively short payback period demonstrates the system's potential to provide a cost-effective cooling solution, particularly beneficial for residents facing financial constraints with cooling expenses. After this period, occupants gain access to a largely free cooling source, offering a more affordable and sustainable option compared to traditional A/Cs, with minimal site disturbance required for installation.

Table 1. Total anticipated cost associated with the proposed system

Cost Subsection			**Cost**	
	Material	Total Cost	Material	Total Cost
Materials	PMP lightweight concrete mix [28]	$0.88\ m^3 \times \$250/m^3 = \$220$	Insulation (R-13 fiberglass roll) [29]	$12.1\ m^2 \times \$6.3/m^3 = \$76.2$
	Clear float glass [30]	$20.7\ m^2 \times\ \$5.0/m^2 = \$103.5$	Fir Lumber for other parts (water container, fan container, base, etc.) [31]	$2.1\ m^3 \times \$35.2/m^3 = \$73.9$
	SolarSpace 535W solar panel [32]	$55.3	Water (assuming the whole volume needs to be replaced three times a day.) [33]	$3 \times 0.5\ m^3 \times \$1.25/m^3 = \$1.9$
	Fan [34]	$31.3	Rotatable base's electrical and mechanical devices [35]	$82.6
	Total material cost		$644.7	
Construction & Transportation	Designing analysis (discussed in From Concept to Installation section, part 2)		$1\ person \times 1day \times \$100/day.person = \$100$	
	Construction [36]		$40\ m^2 \times \$2.1/m^2 = \$80.2$	
	Transportation (200 miles from the manufacturer to the site is assumed.) [37]		$(200\ miles \times \$2.1/mile)\ /\ 20 = \$21$ (Assuming 20 systems can be transported each time.)	
Installation	Assembly and installation (discussed in From Concept to Installation section, part 3)		$2\ person \times 1day \times \$100/day.person = \$200$	

Potential Adaptation Barrier. A primary adaptation barrier for this system, and a concern for both homeowners and manufacturers, is the additional load it places on a building's existing structure when installed on the roof. This challenge can be addressed through the following potential solutions: 1) Use of Lightweight Materials. Since the system does not bear any structural load, it can be constructed from lightweight materials without compromising its functionality, which minimizes the added weight on the building. 2) Additional Structural Support. If necessary, supplementary structural components alongside the ceiling and walls can be added to transfer the load directly to the ground or the building's foundation, reducing the impact on the existing structure. However, this solution will add to the final cost and should be factored into the project budget.

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